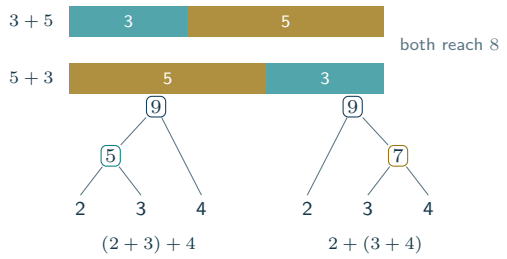
ALGEBRA LADDER 1 OF 9

Operations and their laws

An operation turns two numbers into one, and two laws ask whether the order and the grouping matter.

1 / THE RULE AND THE PICTURE

Commutative means the order does not matter, and associative means the grouping does not matter



Put two lengths end to end and the total is the same whichever comes first, which is the commutative law, and add three numbers in two steps and the answer is the same whichever pair is added first, which is the associative law:

$$a \circ b = b \circ a, \qquad (a \circ b) \circ c = a \circ (b \circ c)$$

The symbol $\circ$ stands for any operation, so each law is a question to ask of an operation, and a few operations pass and many fail.

2 / WHICH OPERATIONS PASS

Addition and multiplication pass both tests, and subtraction, division and powers fail both

operation	laws that hold	checking with numbers
$a + b$	commutative and associative	$3 + 5 = 5 + 3$ and $(2 + 3) + 4 = 2 + (3 + 4) = 9$
$a \times b$	commutative and associative	$6 \times 7 = 7 \times 6$ and $(2 \times 3) \times 4 = 2 \times (3 \times 4) = 24$
$a - b$	neither law	$8 - 3 = 5$ but $3 - 8 = -5$; $(10 - 4) - 3 = 3$ but $10 - (4 - 3) = 9$
$a \div b$	neither law	$12 \div 4 = 3$ but $4 \div 12 = \frac{1}{3}$; $(24 \div 6) \div 2 = 2$ but $24 \div (6 \div 2) = 8$
a^b	neither law	$2^3 = 8$ but $3^2 = 9$; $(2^3)^2 = 64$ but $2^{(3^2)} = 512$
average of a, b	commutative only	$\frac{2+4}{2} = \frac{4+2}{2}$; but $\text{avg}(\text{avg}(2,4),6) = 4.5$ and $\text{avg}(2,\text{avg}(4,6)) = 3.5$

3 / WHY THE LAWS ARE USEFUL

Reordering and regrouping make hard sums easy

To find $25 \times 17 \times 4$ the commutative law lets us swap to $25 \times 4 \times 17$, and the associative law groups $(25 \times 4) \times 17 = 100 \times 17 = 1700$. To find $98 + 47 + 2$ we can add $98 + 2 = 100$ first and then 47 to get 147. In both sums the two numbers that combine to something friendly are brought together, which is allowed only because addition and multiplication obey the laws, and the same moves would give wrong answers for $98 - 47 - 2$ or $24 \div 6 \div 2$.

4 / A CONVENTION IS DIFFERENT FROM A LAW

Brackets and the order of operations tell us how to read a line of symbols

Reading $2 + 3 \times 4$ as 14 and not 20 is an agreement that multiplication is done before addition, made so that a line of symbols has one meaning, and brackets override it: $(2 + 3) \times 4 = 20$.
The agreement and the laws work together, since the next sheet shows that the rule "multiply before adding" is exactly what lets us split 7×13 into $7 \times 10 + 7 \times 3$ without brackets.

5 / A LAW THAT FAILS IN DAILY LIFE

Doing one thing and then another can matter

Putting on socks and then shoes is a different action from putting on shoes and then socks, so the operation "do this then that" is not commutative. It is still associative, because doing three tasks in order gives the same result however we group them in our heads, and this pair of properties (order matters and grouping does not) returns for turns, flips and matrices later in the ladder.

6 / TRY THESE

Decide which laws hold and prove a failure with one example

(a) Is $a \circ b = a + 2b$ commutative? Take $a = 1, b = 2$: $1 + 4 = 5$ but $2 + 2 = 4$, so it fails.
(b) Is "the larger of a and b" associative? $\max(\max(2,5),3) = 5 = \max(2,\max(5,3))$, and it holds for every triple.
(c) Mixing red and blue paint into purple has the same result in either order, so we say mixing is commutative. One counterexample breaks a law, and many matching examples can never prove one.

ASIDE / A LAW IS A PROMISE

A law has to hold for every number, so a single counterexample ends it while any number of agreeing examples leaves it unproved.

LINKS TO OTHER SHEETS

The next sheet uses these laws to multiply by splitting numbers into parts; the modular clock in the number theory ladder is a world where both laws still hold.

Convention. $\circ$ is a placeholder for any operation; $\max$ is the larger of two numbers; avg is the average.